Estimation of the Impact of Dimensionality Reduction of the Feature Space on the Efficiency of Movements Classification based on Surface Electromyography

Ivan V. Kuznetsov
Department of Computer Science and Computer Graphics
Far Eastern State Transport University
Khabarovsk, Russia
ORCID: 0000-0002-9110-2816

Yulia V. Ponomarchuk
Faculty of Control, Automation and Telecommunications
Far Eastern State Transport University
Khabarovsk, Russia
ORCID: 0000-0002-2531-030X

***Abstract*— One of the actual tasks of movement classification based on electromyography is the choice of the most informative set of features characterizing the movements. Increasing the number of features may decrease the learning performance of classifiers and reduce the efficiency of gesture classification. This paper is devoted to study the possibilities of improving the movement's classification efficiency by using the feature space dimensionality reduction methods. Classifiers based on Support Vector Machines (SVM), Decision Trees (DT), Random Forest (RF) and K-nearest neighbors (KNN) were considered, and Principal Component Analysis method (PCA) was used to dimensionality reduction. 3% improvement in classification performance was achieved using PCA and KNN.**

Keywords*— *movement classification, electromyography, principal component analysis, dimensionality reduction, machine learning, SVM, DT, RF, KNN

I. Introduction

Electromyography (EMG) is a method of recording the difference of potentials arising from the activation of skeletal muscles. It is one of the most promising sources of control signal for developing human-machine interfaces (HMI). To solve this task, EMG-signal is collected by surface electrodes and is the result of interference of multiple EMG signals of individual motor units (MU).

To identify gestures that can act as commands for end devices, a large number of characteristics can be extracted from the EMG signal to be used as features of gesture classes and used to train classifiers based on machine learning and deep learning methods.

The features that can be extracted from EMG signal belong to Time Domain (TD), Frequency Domain (FD) or Time-Frequency Domain (TFD). TD features are characterized by their ease of computation and therefore are common used [1]. The disadvantage of TD features is the variability of their statistical properties due to the non-stationarity of the EMG signal [2, 3]. In turn, the use of FD features is complicated by the high sensitivity of the EMG signal to noise [4].

TFD features analysis is a promising direction of research: it is considered that they allow to overcome the frequency and time domains limitations [4, 5], but their disadvantage is the high dimensionality of the feature space, which limits the quality of classifier training. At the same time, it is impossible to guarantee that all extracted TFD features are sufficiently informative.

In order to improve the efficiency of gesture classification, dimensionality reduction methods are used, in particular, the Principal Component Analysis. The purpose of this study is to investigate the classification performance when PCA is used compared to classification without it. The assumption of increased classification performance when PCA is used is considered.

II. Materials and methods

A. Classsifiers to be evaluated

To validate the hypothesis, a comparative analysis of the efficiency of the four sets of classifiers was performed. Each set contained four classifiers based on the following methods:

- SVM
- DT
- RF
- KNN

The same architectures were used for all sets of classifiers. The first set was based on TD features. The second group of classifiers additionally used TFD features. For the third and fourth groups of classifiers, that implemented in a similar way, PCA was additionally used to reduce the dimensionality of the feature space.

Before training the classifiers, the dataset was normalized using l_2 norm and standardized. The Scikit-learn library and Python programming language were used to perform the calculations.

B. Dataset used in experiment

The NinaPro DB5 dataset was used in the experiment. This is an EMG signal dataset obtained from 10 subjects using two Thalmic Labs Myo armbands [6]. The movements in the dataset are described using 16-channel electromyogram data (eight channels per wristband). The experiment did not use any other information (e.g., motion kinematics data) provided in the set to train the classifiers.

In total, the dataset contains information about fifty two movements, out of which 12 movements are selected for classification, which are shown in Fig. 1.

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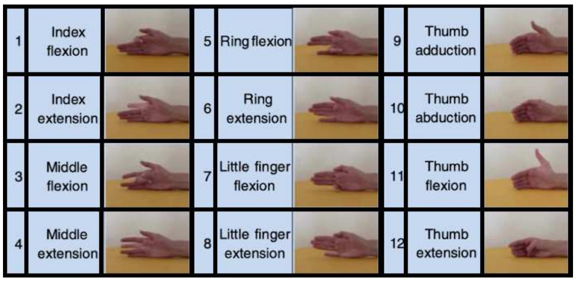

Fig. 1. Movements selected for the experiment

During the experiment, it was accepted that the information presented in the dataset did not require any additional preprocessing.

C. Time Domain Features

The following TD features were used to describe the movements: Variance, Zero Crossing (ZC), Difference Absolute Standard Deviation Value (DASDV).

Zero Crossing represents the number of crossings of the zero level by the EMG signal. The methods of ZC calculation are discussed in [7], where it is indicated that during the analysis it is required to take into account the influence of noise. The following expressions were used for calculation:

$$ZC = \sum_{i=1}^{N-1} ZC(x_i), \quad T = 4\left(\frac{1}{10}\sum_{j=1}^{10} x_j\right),$$

$$ZC(x_i) = \begin{cases} 1, (x_i \geq \mathrm{T}) \text{ and } (x_{i+1} < T) \\ 1, (x_i < \mathrm{T}) \text{ and } (x_{i+1} \geq T) \\ 0, \text{otherwise} \end{cases}$$

which T is the threshold coefficient.

Variance, according to Spiewak [8], describes the power of the EMG signal. The sign can be calculated as follows:

$$VAR = \frac{1}{N-1}\sum_{i=1}^{N}\sqrt{x^2}$$

The Mean Absolute Value feature was used in [1]. It is noted in [8] that MAV is a method of defining and measuring the level of muscle contraction, which can be calculated using the following expression:

$$MAV = \frac{1}{N}\sum_{i=1}^{N}|x_i|$$

The Difference Absolute Standard Deviation Value feature was used in the Phimyomark study [9]. It represents the absolute standard deviation for the difference of neighbouring values:

$$DASDV = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}(x_{i+1} - x_i)^2}$$

Presented features were used in all groups of classifiers.

D. Time-Frequency Domain Features

The Discrete Wavelet Transform with four-level decomposition was applied to EMG signal data to compute the features of the frequency-time domain. The fifth order Daubechies wavelet was used as the mother wavelet.

Then, Variance, Root Mean Square (RMS) and Energy Discrete Wavelet Transform (EDWT) signs are calculated for the obtained coefficients [10]:

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$$

$$EDWT = \sum_{i=1}^{N} c_i^2$$

which c_i is the coefficient obtained as a result of the wavelet transform.

The presented features were used in the third and fourth group classifiers.

III. Results and Discussion

Recall and Precision efficiency metrics were calculated to assess performance:

$$Precision = \frac{TP}{TP + FP}, Recall = \frac{TP}{TP + FN}$$

with TP is the percentage of objects correctly assigned to a given class, TN – the percentage of objects correctly not assigned to a class, FN is the percentage of objects incorrectly not assigned to a given class, FP is the percentage of objects incorrectly assigned to a given class.

The classes in the dataset are not balanced: there are significantly more objects describing the resting state than gesture objects. All classes, despite the frequency of occurrence of the corresponding objects, have the same importance – when designing HMIs, one command cannot be considered more important than another. This needs to be considered when evaluating the classifiers performance.

The classifiers performance metrics are presented in Table 1. Macro-averaging was used to calculate the metrics.

TABLE I. Classifier Performance Metrics

	SVM	**DT**	**RF**	**KNN**
Precision(TD)	0.71	0.56	0.77	0.76
Recall(TD)	0.51	0.57	0.70	0.73
Precision(TD+PCA)	0.72	0.58	0.80	0.78
Recall(TD+PCA)	0.52	0.59	0.67	0.77
Precision(TD+TFD)	0.78	0.63	0.81	0.82
Recall(TD+TFD)	0.76	0.62	0.80	0.82
Precision(TD+TFD+PCA)	0.78	0.66	0.83	0.85
Recall(TD+TFD+PCA)	0.76	0.63	0.81	0.85

The confusion matrices of the first and third groups of classifiers for KNN are presented in Fig. 2. It should be noted that the classifiers incorrectly recognize thumb movements, little finger and ring finger movements, as well as index finger movements when TD features are used. It can be assumed that this is due to the use of closely located motor units: due to the fact that in the case of surface EMG the

EMG-signal represents the interference of the EMG-signals of individual MUs, classifiers are unable to distinguish movements that use closely located MUs based on only TD features. KNN and RF showed the best efficiency. The effectiveness of RF was significantly higher than that of DT. This suggests that the use of ensemble methods for classifying movements can improve the efficiency of solving the classification problem.

The use of PCA leads to an increase in the overall classification efficiency. The classification efficiency does not change for the thumb, decreases for the ring finger, and increases for the little and index fingers. This allows us to state that when using PCA, some useful information is lost, and also to conclude that it is necessary to clarify the location of the reading electrodes in order to increase the proportion of successfully classified thumb movements.

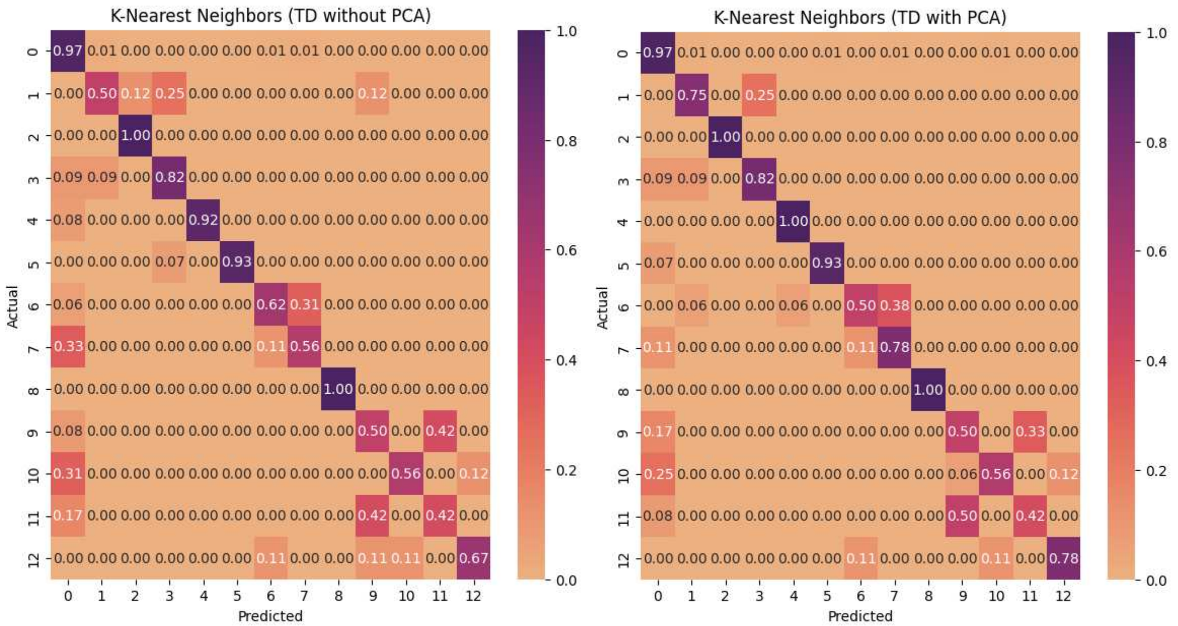

Fig. 2. Confusion matrices of the first and second group KNN classifiers

When using TFD features in addition to TD for all classifiers, an increase in classification efficiency is found. With the additional use of PCA for the SVM-based classifier, the Precision and Recall values did not change.

DT and RF based classifiers significantly increased the performance relative to their counterparts using only TD features. When using PCA for classifiers with TFD features, an increase in classification efficiency of up to 3% is achieved.

The best results were achieved for KNN with 85% for Precision and Recall using TFD features and PCA. The error matrix of this classifier as well as its analogue that does not use PCA is shown in Fig. 3. It can be seen that the increase in performance is achieved by reducing the percentage of incorrect classification of the index finger, ring finger and little finger movements. The success rate of classification of thumb gestures, changed insignificantly.

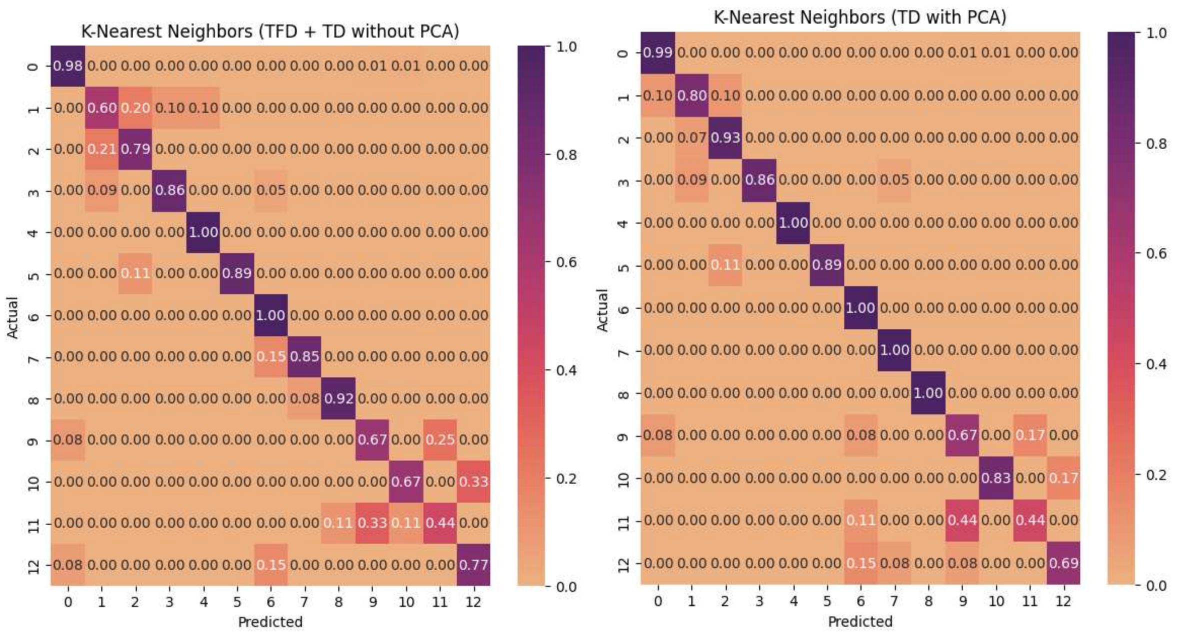

Fig. 3. Confusion Matrices of the third and fourth group KNN classsifiers

The increase in overall efficiency for KNN when using TFD features and PCA is about 3% relative to its analogue that does not use PCA. This allows us to conclude that the use of PCA is acceptable for increasing the efficiency of movement classification.

IV. Conclusion

During the experiment a good success rate in classifying the movements of the little finger, ring finger, index finger and middle finger was achieved. For the thumb, an increase of the classification efficiency was achieved, but the challenge of improving gesture recognition efficiency remains open. As one of the solutions, it is proposed to superimpose the sensing electrodes taking into account the location of the MUs responsible for the considered movements.

The use of PCA increased the success rate of classified movements for RF, DT and KNN based classifiers. This paper considers only one of many possible feature sets, which raises the question of finding more effective sets.

At the same time, it is impossible to guarantee that PCA will be effective for new feature sets to the same extent. This makes it urgent to search for other effective dimensionality reduction methods and compare the results of their application. It is expected that the methods of independent component analysis, as well as T-distributed Stochastic Neighbor Embedding will be effective.

Based on the study, gestures with high classification performance can also be selected for HMI development.

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